



EE4402 Project

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Video Link:

<https://share.weiyun.com/B1DvNPfK>



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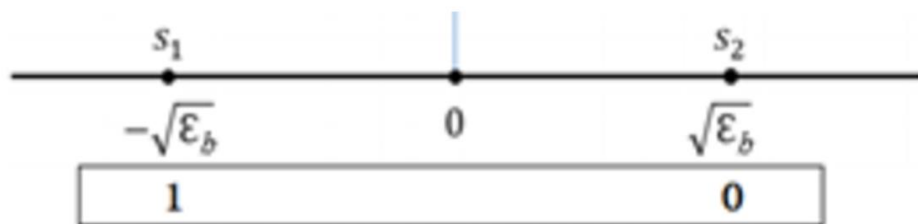
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Task 1

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Task 2

- **Task 1** Assume all symbols are equiprobable. Denote the average bit energy by E_b . Derive and simulate the bit error rate (BER) for PAM when the Gray mapping is applied.
- **Task 1 (a)** If 2-PAM and 4-PAM have the same average bit energy E_b , indicate the symbols in vector form for 2-PAM and 4-PAM using E_b .



Assume $S_{ml}(t) = A_m p(t)$ $A_m = 2m - 1 - M$ $1 \leq m \leq M = 2^k$

Denote $\varepsilon_p = \int_{-\infty}^{\infty} p(t)^2 dt \rightarrow \varepsilon_{ml} = A_m^2 \varepsilon_p = E_b$

PAM has dimension one $\Phi_l(t) = \frac{p(t)}{\sqrt{\varepsilon_p}}$

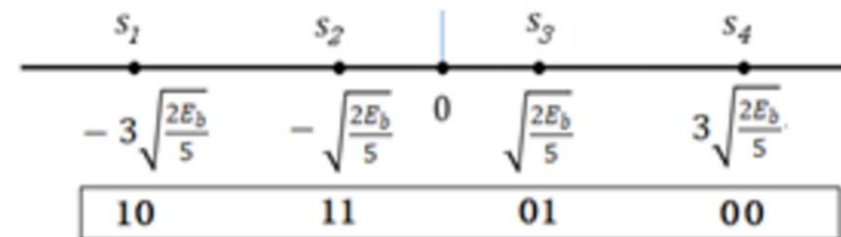
2PAM(M=2):

$$\varepsilon_{ml} = A_m^2 \varepsilon_p = \varepsilon_p$$

$$\varepsilon_{avgl} = \varepsilon_p \quad \varepsilon_{bavgl} = \frac{\varepsilon_{avgl}}{\log_2 M} = \varepsilon_p = E_b$$

$$S_1 = \langle S_{1l}(t), \Phi_l(t) \rangle = -\sqrt{\varepsilon_b} = -\sqrt{E_b} \quad S_2 = \langle S_{2l}(t), \Phi_l(t) \rangle = \sqrt{E_b}$$

Vector form: $S_1 = (-\sqrt{E_b}, 0)$ $S_2 = (\sqrt{E_b}, 0)$



4PAM(M=4):

$$A = \pm 1, \pm 3 \quad \varepsilon_{ml} = A_m^2 \varepsilon_p = 1, 9, 9, 1 \varepsilon_p$$

$$\varepsilon_{avgl} = 5\varepsilon_p \quad \varepsilon_{bavgl} = \frac{\varepsilon_{avgl}}{\log_2 M} = \frac{5}{2} \varepsilon_p = E_b$$

$$S_1 = \langle S_{1l}(t), \Phi_l(t) \rangle = \int_{-\infty}^{\infty} -3p(t) * \frac{p(t)}{\sqrt{\varepsilon_p}} dt = -3\sqrt{\varepsilon_p} = -3\sqrt{\frac{2E_b}{5}}$$

$$S_2 = \langle S_{2l}(t), \Phi_l(t) \rangle = -\sqrt{\frac{2E_b}{5}}$$

$$S_3 = \langle S_{3l}(t), \Phi_l(t) \rangle = \sqrt{\frac{2E_b}{5}} \quad S_4 = \langle S_{4l}(t), \Phi_l(t) \rangle = 3\sqrt{\frac{2E_b}{5}}$$

Vector form: $S_1 = (-3\sqrt{\frac{2E_b}{5}}, 0)$ $S_2 = (-\sqrt{\frac{2E_b}{5}}, 0)$ $S_3 = (\sqrt{\frac{2E_b}{5}}, 0)$ $S_4 = (3\sqrt{\frac{2E_b}{5}}, 0)$

- Task 1 (b)** Derive the exact BER for 2-PAM and 4-PAM in terms of the signal-to-noise ratio (SNR) defined as $\gamma_b = \frac{E_b}{N_0}$ when the Gray mapping is applied.

2PAM:

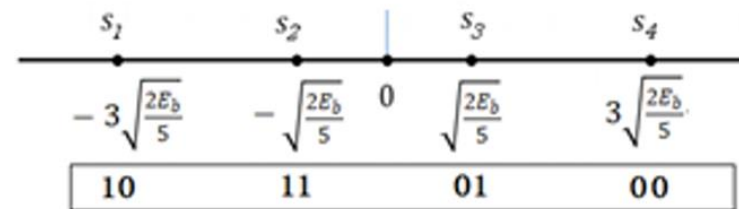


SER:

$$\begin{aligned}
 P_e &= \int_{-\infty}^0 P(s_2, r) dr + \int_0^{\infty} P(s_1, r) dr \\
 &= \int_{-\infty}^0 P(r|s_2)P(s_2)dr + \int_0^{\infty} P(r|s_1)P(s_1)dr \\
 &= \frac{1}{2} \left(\int_{-\infty}^0 \frac{1}{\sqrt{\pi N_0}} e^{-\frac{(r - \sqrt{E_b})^2}{N_0}} dr + \int_0^{\infty} \frac{1}{\sqrt{\pi N_0}} e^{-\frac{(r + \sqrt{E_b})^2}{N_0}} dr \right) \\
 &= Q \left(\sqrt{\frac{2E_b}{N_0}} \right) = Q \left(\sqrt{\frac{2E_b}{N_0}} \right) = Q(\sqrt{2\gamma_b})
 \end{aligned}$$

Since there are 2 signals, **BER = SER = $Q(\sqrt{2\gamma_b})$.**

4PAM:



$$s_1 = (-3\sqrt{\frac{2E_b}{5}}, 0) \quad s_2 = (-\sqrt{\frac{2E_b}{5}}, 0) \quad s_3 = (\sqrt{\frac{2E_b}{5}}, 0) \quad s_4 = (3\sqrt{\frac{2E_b}{5}}, 0)$$

$$\text{Assume that } A = \sqrt{\frac{2E_b}{5}}$$

$$\text{Therefore, } s_1 = (-3A, 0) \quad s_2 = (-A, 0) \quad s_3 = (A, 0) \quad s_4 = (3A, 0)$$

BER:

$$P_{e|s_1} = \frac{1}{2} P_{s_2|s_1} + P_{s_3|s_1} + \frac{1}{2} P_{s_4|s_1} \quad P_{e|s_2} = \frac{1}{2} P_{s_1|s_2} + \frac{1}{2} P_{s_3|s_2} + P_{s_4|s_2}$$

s_1 and s_4 are symmetric, s_2 and s_3 are symmetric.

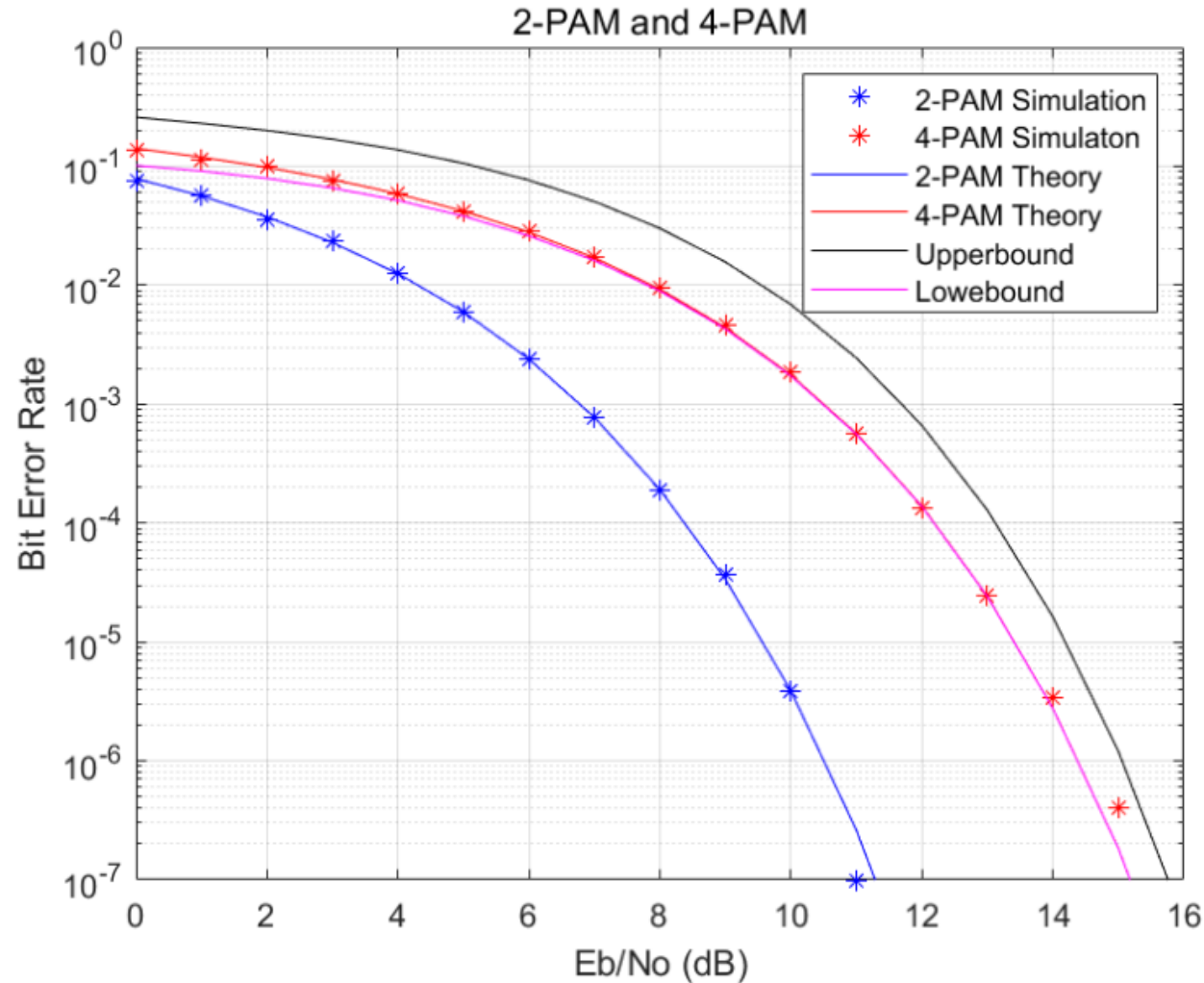
$$\text{Therefore, } P_{e|s_1} = P_{e|s_4}, P_{e|s_2} = P_{e|s_3}.$$

$$\text{As } A = \sqrt{\frac{2E_b}{5}} \quad \gamma_b = \frac{E_b}{N_0}$$

$$P_e = P_{e|s_1} + P_{e|s_2} + P_{e|s_3} + P_{e|s_4} = 2P_{e|s_1} + 2P_{e|s_2}$$

$$= \frac{1}{4} \left[3Q \left(2\sqrt{\frac{\gamma_b}{5}} \right) + 2Q \left(6\sqrt{\frac{\gamma_b}{5}} \right) - Q \left(10\sqrt{\frac{\gamma_b}{5}} \right) \right]$$

- **Task1 (c):** Simulate 2-PAM and 4-PAM modulation and detection in baseband with AWGN to find the experimental BER using Matlab.



- **Task1(d):** Derive an upper bound and a lower bound for the 4PAM BER

From the tutorial we have proved that

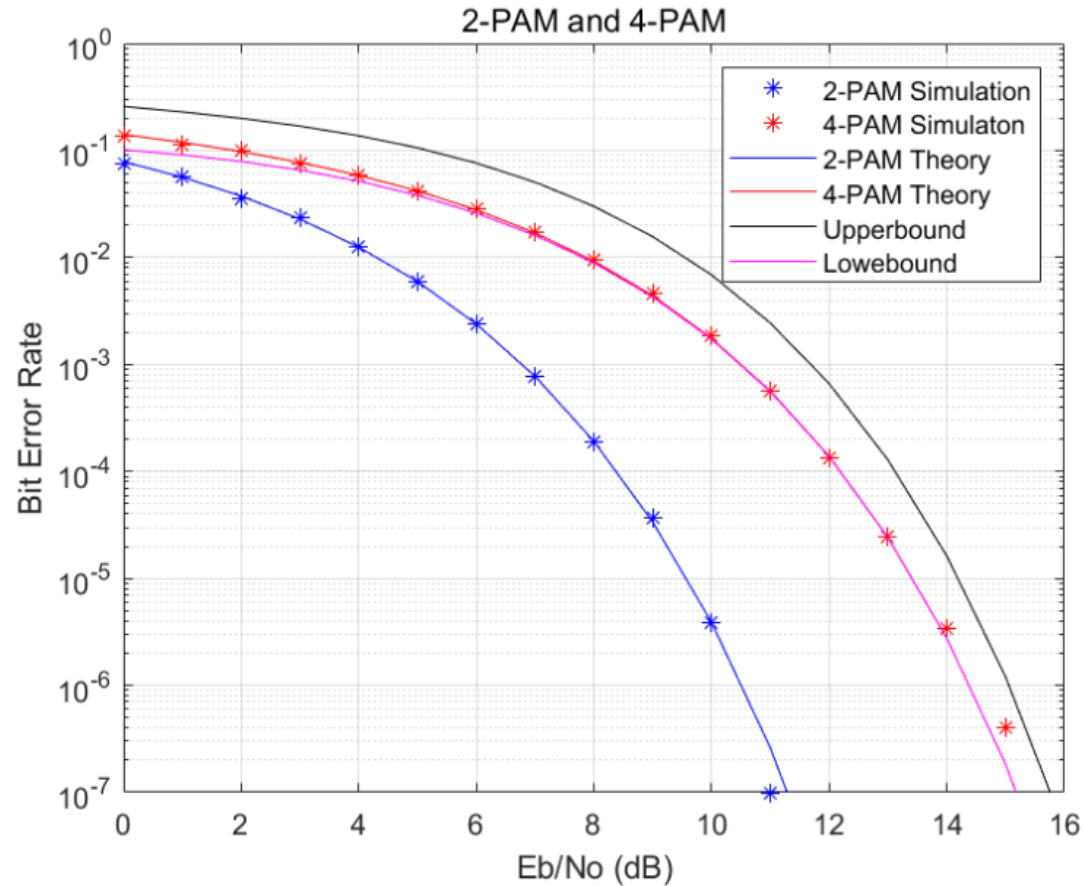
$$\frac{x}{\sqrt{2\pi}(1+x^2)} e^{-\frac{x^2}{2}} < Q(x) \leq \frac{1}{2} e^{-\frac{x^2}{2}}$$

And we have

$$P_e = \frac{1}{4} \left[3Q\left(2\sqrt{\frac{\gamma_b}{5}}\right) + 2Q\left(2\sqrt{\frac{\gamma_b}{5}}\right) - Q\left(2\sqrt{\frac{\gamma_b}{5}}\right) \right]$$

Replace $Q(x)$ in inequality with $Q\left(2\sqrt{\frac{\gamma_b}{5}}\right)$ $Q\left(6\sqrt{\frac{\gamma_b}{5}}\right)$ $Q\left(10\sqrt{\frac{\gamma_b}{5}}\right)$ so we can get the upper bound and the lower bound

•**Task1 (e):** Plot the exact BER vs. SNR, the upper bound, and the lower bound in the same plot and comment on the results.



simulation result

Comments on the result:

1. At the same signal-to-noise ratio, we can see that the BER of 2PAM is lower than that of 4PAM, indicating better modulation performance.

2. The bounds we derived meet the simulation requirements.

- Task2 (a) Sketch the constellation of the QPSK.**

Assume $S_m(t) = \text{Re} \left[p(t) e^{j(\theta_0 + \frac{2\pi(m-1)}{M})} e^{j2\pi f_c t} \right] = p(t) \cos(2\pi f_c t + \theta_0 + \frac{2\pi(m-1)}{M})$

Denote $\varepsilon_p = \int_{-\infty}^{\infty} p(t)^2 dt \rightarrow \varepsilon_{avg} = \varepsilon_m = \frac{\varepsilon_p}{2} \quad \varepsilon_{bavg} = \frac{\varepsilon_{avg}}{\log_2 M} = \frac{\varepsilon_p}{4} = E_b$

PAM has dimension two

$$\left\{ \Phi_1(t) = \sqrt{\frac{2}{\varepsilon_p}} p(t) \cos(2\pi f_c t), \Phi_2(t) = \sqrt{\frac{2}{\varepsilon_p}} p(t) \sin 2\pi f_c t \right\}$$

GSO procedure 

$$\begin{aligned} S_{11} = \langle S_1(t), \Phi_1(t) \rangle &= \int_{-\infty}^{\infty} p(t) \cos(2\pi f_c t + \theta_0) * \sqrt{\frac{2}{\varepsilon_p}} p(t) \cos(2\pi f_c t) dt = \sqrt{\frac{\varepsilon_p}{2}} \\ &= \sqrt{2E_b} \end{aligned}$$

$$S_{12} = \langle S_1(t), \Phi_2(t) \rangle = 0$$

$$S_{21} = \langle S_2(t), \Phi_1(t) \rangle = 0 \quad S_{22} = \langle S_2(t), \Phi_2(t) \rangle = \sqrt{2E_b}$$

$$S_{31} = \langle S_3(t), \Phi_1(t) \rangle = -\sqrt{2E_b} \quad S_{32} = \langle S_3(t), \Phi_2(t) \rangle = 0$$

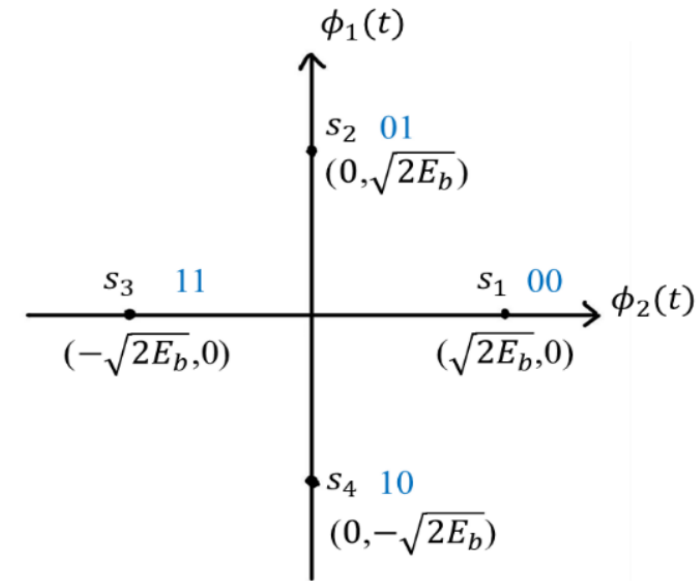
$$S_{41} = \langle S_4(t), \Phi_1(t) \rangle = 0 \quad S_{42} = \langle S_4(t), \Phi_2(t) \rangle = -\sqrt{2E_b}$$

•**Task2 (b)** Mark the symbol vector forms (in terms of E_b) and the corresponding Gray coding.

Then the symbol vector forms of QPSK signals in terms of E_b and the corresponding Gray coding are

signal	symbol vector form	Gray code
S1	$(\sqrt{2E_b}, 0)$	00
S2	$(0, \sqrt{2E_b})$	01
S3	$(-\sqrt{2E_b}, 0)$	11
S4	$(0, -\sqrt{2E_b})$	10

The constellation of the QPSK is



•**Task2 (c):** Derive the exact BER and express it in terms of the bit SNR when the Gray mapping is applied. Derive a upper bound and a lower bound for the QPSK BER in terms of SNR.

The SER of the QPSK signals is

$$P_e = \frac{1}{M} \sum_{i=1}^4 P(e|S_i) = P(e|S_i) = \sum_{\substack{1 \leq k \leq M \\ k \neq i}} \int_{D_k} P(\mathbf{r}|s_i) d\mathbf{r}$$

We can see s_1 is sent by detected as any s_k for any $k \neq 1$

For pair-wise SER, $P_{m \rightarrow n} = Q\left(\sqrt{\frac{d_{mn}^2}{2N_0}}\right)$, s_1 is affected by $s_{2,3,4}$, $d_{12} = d_{14} =$

$$2\sqrt{E_b}, d_{13} = 2\sqrt{2E_b}$$

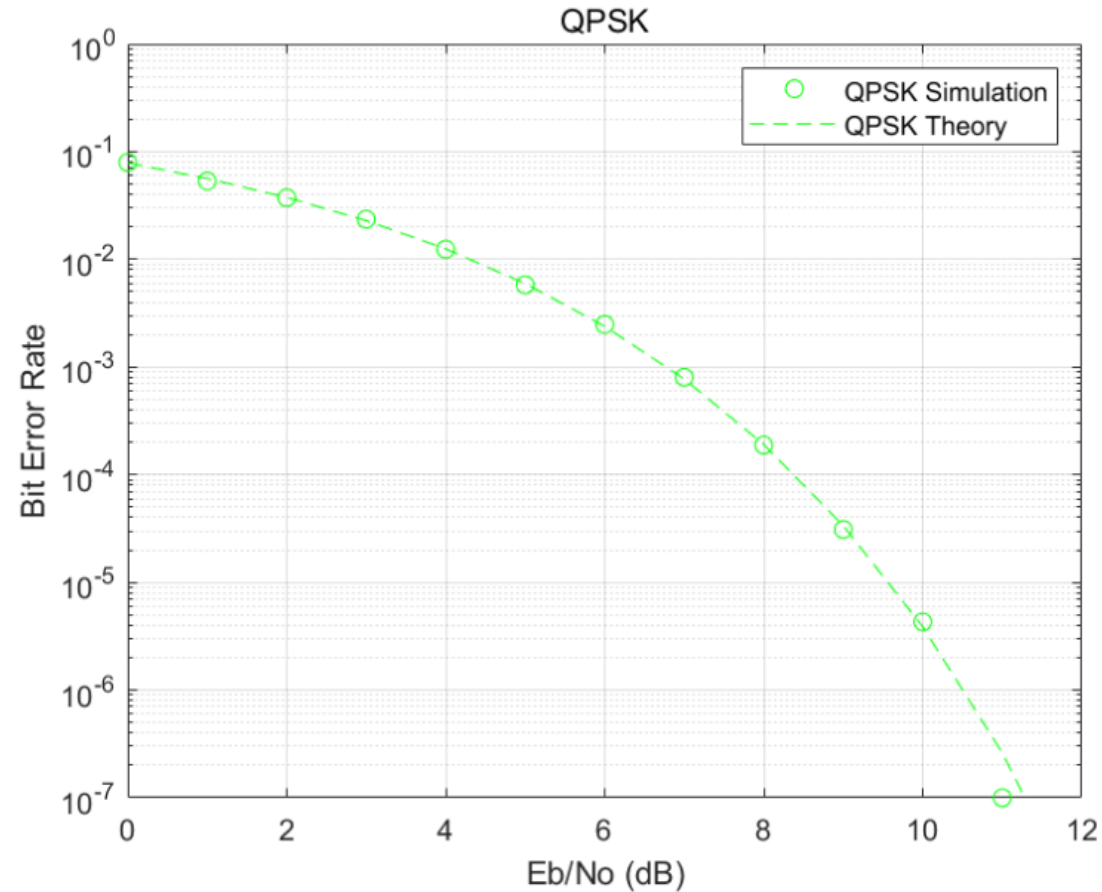
$$P_{1 \rightarrow 2} = P_{1 \rightarrow 4} = Q\left(\sqrt{\frac{4E_b}{2N_0}}\right) = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

Then we can get BER of the QPSK signals

$$\begin{aligned} P_b &= \frac{1}{M} \sum_{i=1}^4 P(b|S_i) = P(b|S_i) = P(b|S_1) \\ &= \frac{1}{2} \int_{D_2} P(\mathbf{r}|s_1) d\mathbf{r} + \int_{D_3} P(\mathbf{r}|s_1) d\mathbf{r} + \frac{1}{2} \int_{D_4} P(\mathbf{r}|s_1) d\mathbf{r} \\ &= \int_{D_2} P(\mathbf{r}|s_1) d\mathbf{r} + \int_{D_3} P(\mathbf{r}|s_1) d\mathbf{r} = P_{1 \rightarrow 2} = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) \end{aligned}$$

Because $\gamma_b = \frac{E_b}{N_0}$ $P_b = Q(\sqrt{2\gamma_b})$

•**Task2 (d)(e):** Simulate QPSK modulation and detection in baseband with AWGN to obtain experimental BER using Matlab; Plot the exact BER vs SNR and comment on the result.



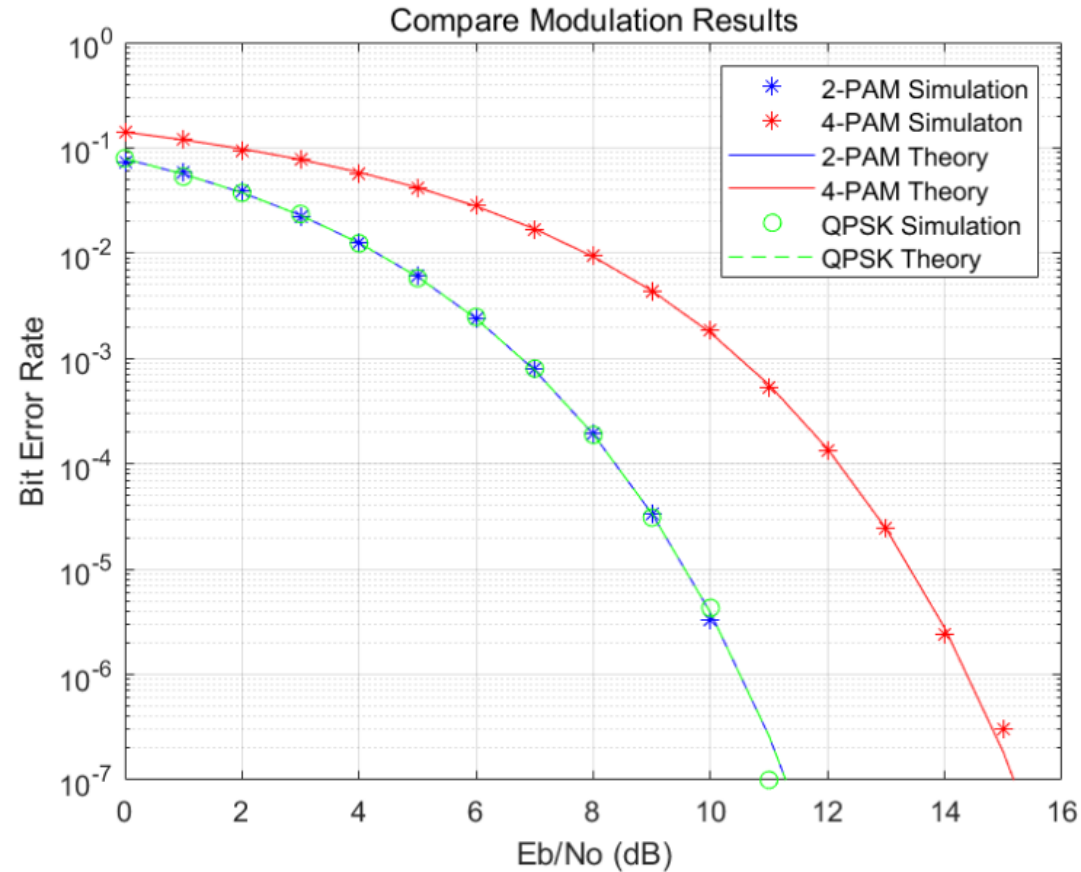
simulation result

comment on the result:

From the result, we can see that

1. BER decreases as the SNR gradually increases.
2. Theoretical bit error rate is the same as actual bit error rate.

➤ **Task2 (f):** Compare the simulation results in Task 1 and Task 2.



Analysis of Results

1.The relationship of BER among the three modulation modes can be obtained at the same SNR:

BER: 4PAM > QPSK = 2PAM

2.At the same BER, 4PAM requires a higher SNR, while 2PAM and QPSK have higher efficiency.



THANKS